Emotion Detection in Multilingual Social Media Text Using NLP

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1. What you will deliver

Requirement	What you do	Where
Real-world dataset, documented	Source, size, languages, label distribution, length	Section 3, `results/dataset_stats.csv`
Preprocessing and features	Social-media cleaning, word and char TF-IDF, XLM-R subwords	Section 5, `common.py`
At least two techniques	Baseline TF-IDF + Logistic Regression, improved TF-IDF (word+char), fine-tuned XLM-RoBERTa	`baseline.py`, `xlmr.py`
Metrics	Precision, Recall, F1 (micro and macro), subset accuracy, Hamming loss	Section 7
Tables and graphs	Comparison table, per-language and per-emotion charts	`report.py`
Error analysis and limitations	FP/FN per emotion, per language, length bins, confusion pairs, examples	Sections 9-10
Improvement and research direction	Derived from your own error analysis	Section 10
Source code, dataset details, methods, results, references	Folder layout and checklist	Sections 4, 13, 14

Important: dataset details below are from memory. Open each dataset page and confirm sizes, language codes, licence and label set before you write them in your report.

2. Problem definition

Task. Given a social media post written in language L, predict which emotions it expresses from {anger, fear, joy, sadness, surprise}. A post can carry several emotions or none, so this is multi-label classification (an empty label set means no emotion or neutral).

Why it is hard.

- Slang, abbreviations, emojis, hashtags and elongated words (sooo good)
- Code-mixing and transliteration (Hinglish: mujhe bahut gussa aa raha hai)
- Sarcasm, negation and missing context in very short text
- Class imbalance and cultural differences in how emotion is expressed and annotated
- Low-resource languages with little labelled data

Objectives.

1. Build a reproducible pipeline for multilingual emotion detection.
2. Compare classical (TF-IDF) and transformer (XLM-R) approaches.
3. Study multilingual training and cross-lingual (zero-shot) transfer.
4. Analyse errors and propose a research direction.

Research questions.

- RQ1: Does fine-tuned XLM-R outperform TF-IDF models, and by how much?
- RQ2: Does training on several languages beat training on one language only?
- RQ3: How well does a model trained on English transfer to another language without target-language training data?
- RQ4: Which emotions and languages are hardest, and why?

Hypotheses (state, then test). H1: XLM-R > TF-IDF in macro-F1. H2: char n-grams help over word-only TF-IDF, especially in morphologically rich or non-space-delimited languages. H3: zero-shot transfer performs below in-language training.

3. Dataset

Primary: BRIGHTER / SemEval-2025 Task 11 (Track A)

- Multilingual, human-annotated emotion dataset covering about 28 languages (social media and other sources), roughly 100k instances in total.
- Multi-label: anger, fear, joy, sadness, surprise (some languages also have disgust). Intensity labels exist for Track B.
- Hugging Face id used in the code: `brighter-dataset/BRIGHTER-emotion-categories` (one config per language). Also available from the task GitHub repository.
- Run `python common.py` to print the exact language configs that exist, then choose your subset.
- Suggested subset: 3-5 languages with different scripts and resource levels, for example English (high-resource), Hindi (Devanagari, India), one European language, and one low-resource language. If your preferred language (for example Gujarati) is not covered, say so openly as a limitation.
- The code trains on the `train` split (with 10% held out as validation) and reports final numbers on the official `dev` split. Reason: test labels may not be public. State this in your paper.

Alternatives and extras

Dataset	Languages	Labels	Notes
SemEval-2018 Task 1 (E-c)	English, Arabic, Spanish tweets	11 emotions, multi-label	About 7-11k tweets per language; classic benchmark
XED	About 30 languages, OpenSubtitles	8 Plutchik emotions + neutral	Annotations projected across languages; not social media
GoEmotions	English Reddit	27 emotions + neutral	About 58k comments; English only, good for transfer
dair-ai/emotion	English tweets	6 emotions, single-label	About 20k tweets; easy to start with

To use another dataset, write CSVs named `<lang>_train.csv` and `<lang>_dev.csv` into `data/` with columns `text, anger, fear, joy, sadness, surprise` (0/1). The code reads them automatically.

What to document in the report

- Name, citation, URL, licence
- How the data was collected and annotated (annotators, agreement, label definition)
- Languages, script, size per split per language
- Label distribution, percentage of no-emotion and multi-emotion posts, average length
- Ethics: public, anonymised data; no attempt to identify users
- Your split policy (train/val/held-out)

`baseline.py` creates `results/dataset_stats.csv` with most of this. Put it in the paper as a table.

4. Setup and how to run

Folder layout

```
emotion-project/
  common.py      # loading, cleaning, metrics, saving
  baseline.py    # TF-IDF models (baseline + improved)
  xlmr.py        # fine-tune XLM-RoBERTa
  report.py      # tables, charts, error analysis
  requirements.txt
  data/          # auto-created
  results/       # auto-created
```

Environment. Use Google Colab (free T4 GPU) for XLM-R. TF-IDF runs on any laptop.

```
pip install -r requirements.txt
python common.py                                    # list language configs
python baseline.py --train_langs eng hin deu        # E1, E2
python xlmr.py --train_langs eng hin deu --epochs 4   # E3
python report.py --tag multi                        # tables, charts, error
analysis

# zero-shot transfer (E5)
python baseline.py --train_langs eng --test_langs hin --tag zs_eng2hin
python xlmr.py --train_langs eng --test_langs hin --tag zs_eng2hin
python report.py --tag zs_eng2hin
```

Replace `eng hin deu` with codes returned by `python common.py`.

5. Preprocessing and feature representation

Step	Why
Unicode NFKC normalisation	Unifies visually identical characters across scripts
URLs to `<url>`, mentions to `<user>`	Removes noise and protects privacy
Hashtag symbol removed, word kept	The word often carries the emotion
Elongation squashed (sooooo to soo)	Reduces vocabulary while keeping emphasis signal
Lowercasing, whitespace cleanup	Standard normalisation
Emojis kept	Strong emotion signal; XLM-R tokenizer handles them

Deliberately not done: English stopword removal and stemming. They are language-specific, would damage other languages, and removing words like not or never hurts emotion detection. Explain this choice in the paper.

Feature representations

1. Word TF-IDF (unigrams + bigrams, sublinear tf). Sparse, interpretable, baseline.
2. Character n-gram TF-IDF (2-5, word-boundary). Language-agnostic, robust to spelling variation and transliteration.
3. XLM-R contextual subword representations (SentencePiece vocabulary shared across 100 languages), learned during fine-tuning.

6. Models

ID	Model	Role
M1	Word TF-IDF + One-vs-Rest Logistic Regression	Baseline
M2	Word + char TF-IDF + One-vs-Rest Logistic Regression	Improved classical
M3	Fine-tuned `xlm-roberta-base` with sigmoid output layer	Improved neural

Key design choices

- One-vs-Rest because emotions are not mutually exclusive.
- Class weights (LR) and positive-class weights (XLM-R) to counter imbalance.
- Sigmoid + binary cross-entropy for XLM-R (not softmax).
- Per-emotion decision thresholds tuned on the validation set, then frozen for the test set.
- XLM-R defaults: learning rate 2e-5, batch 16, 4 epochs, max length 128, warmup 10%, best epoch chosen by validation macro-F1.

Optional extensions (pick one if time permits)

- Multilingual BERT or distilled models for speed comparison
- `cardiffnlp/twitter-xlm-roberta-base` (tweet-adapted) or `google/muril-base-cased` (Indian languages) via `--model`
- Ensemble of TF-IDF probabilities and XLM-R probabilities
- Translate-train augmentation for low-resource languages

7. Evaluation

- **Precision, Recall, F1** per emotion; **macro** (average over emotions, treats rare emotions equally; primary metric) and **micro** (pooled counts).
- **Subset accuracy:** fraction of posts where the whole label set is exactly right. Strict, so values are low; explain this.
- **Hamming loss:** fraction of wrong label decisions. Lower is better.
- Report overall, per emotion and per language.

- Protocol: fixed seed, tune thresholds and pick epochs on validation only, touch the held-out set once per model.
- Optional rigor: paired bootstrap confidence interval for the F1 difference between baseline and best model (done in `report.py`). Repeat XLM-R with different `--seed` values and different tags, then report mean and standard deviation yourself.

8. Experiments to run

ID	Experiment	Answers
E1	M1 on all chosen languages	Baseline
E2	M2 on all chosen languages	RQ1, H2
E3	M3 on all chosen languages	RQ1, H1
E4	Train on one language only vs on all (tags `mono_xx`, `multi`)	RQ2
E5	Train English, test on another language (zero-shot)	RQ3, H3
E6	Threshold tuning on vs off (XLM-R JSON stores `overall_at_0.5`)	Design ablation

Fill a table like this for each:

Model	Micro-F1	Macro-F1	Macro-P	Macro-R	Subset Acc	Hamming
M1						
M2						
M3						

Do not invent numbers. Only report what your runs produced.

9. Error analysis

`report.py` produces: false positives and false negatives per emotion, error rate per language, error rate by text length, a matrix of missed emotion vs wrongly predicted emotion, and `errors_sample.csv` with the most confident mistakes.

Then do this manually. Read 50-100 errors from the best model and tag each one:

Category	Example question
Sarcasm / irony	Is the literal meaning opposite to the emotion?
Negation	Did the model ignore not, never?
Code-mixing / transliteration	Is the text in Roman script for a non-Latin language?
Emoji dependence	Does the emotion live only in an emoji?
Missing context	Would you need the previous post?
Ambiguous label	Fear vs sadness vs surprise; would two annotators disagree?
Label noise	Is the gold label arguably wrong?
Language-specific	Idioms, dialect, spelling variants

Report the percentage of errors per category in a small table and quote 3-5 examples with translations.

10. Limitations and future scope

Typical limitations to discuss (confirm with your results)

- Only a subset of languages; results may not generalise to others
- Held-out set is the official dev split, not the hidden test set
- Single run or few seeds, so small gaps may be noise
- Social media text only; no multimodal or conversational context
- Annotation subjectivity limits the ceiling for any model
- XLM-R base is heavy for low-resource deployment

Research directions (choose the one your error analysis supports)

1. Choosing source languages for transfer by language family or script similarity
2. Handling code-mixed and transliterated text
3. Emoji and culture-aware emotion modelling
4. Comparing fine-tuned models with few-shot LLM prompting
5. Translation-based data augmentation for low-resource languages

6. Model distillation for efficient multilingual emotion detection
7. Explainability (LIME, SHAP, attention) to show which words drive predictions
8. Fairness: performance gaps between high- and low-resource languages

11. Research paper guide

Suggested title: Emotion Detection in Multilingual Social Media Text: A Comparative Study of TF-IDF and XLM-RoBERTa Models

Format: IEEE conference template, 6-8 pages.

Section	What to write	Length
Abstract	Problem, data, methods, key result with numbers, takeaway	150-200 words
Introduction	Why multilingual emotion detection matters, challenges, contributions (3 bullets)	1 page
Related work	Emotion models (Ekman, Plutchik), classical methods, transformers, multilingual datasets and shared tasks	1 page, 15-25 refs
Dataset	Table of languages, sizes, label distribution; annotation; split policy	0.75 page
Methodology	Pipeline figure, preprocessing, features, models, training settings	1.5 pages
Experimental setup	Metrics, hardware, hyperparameters, seeds	0.5 page
Results	Main table, per-language chart, per-emotion chart, zero-shot results	1.25 pages
Error analysis and discussion	Categories, examples, answers to RQ1-RQ4	1 page
Limitations	Honest list	0.25 page
Conclusion and future work	Findings and one clear direction	0.5 page
References	IEEE style	

Abstract template: Emotion detection in multilingual social media text is challenging due to [reasons]. We study [N] languages from [dataset] and compare [M1], [M2] and fine-tuned XLM-RoBERTa on multi-label classification of five emotions. XLM-R achieves a macro-F1 of [x] versus [y] for the TF-IDF baseline. Error analysis shows [main error types]. We also evaluate zero-shot transfer from [source] to [target] and find [result]. We discuss limitations and propose [direction].

Figures and tables to include: pipeline diagram, dataset statistics table, main comparison table, per-language bar chart, per-emotion bar chart, confusion-pair heatmap, error category table.

Writing rules: report only what you ran, cite the dataset and every model, include a reproducibility note (seed, versions, code link), write in your own words and run a similarity check before submission.

12. Suggested timeline

Week	Tasks
1	Read 8-10 papers, fix dataset and languages, run `common.py`, dataset statistics
2	Baselines (E1, E2), debug pipeline, first charts
3	XLM-R (E3), mono vs multi (E4), zero-shot (E5)
4	Error analysis, ablation (E6), write paper and report, prepare viva

13. Submission checklist

- Code folder with README (copy Section 4) and `requirements.txt`
- Dataset details, preprocessing description, model settings
- Result tables, charts, `errors_sample.csv`
- Error analysis, limitations, future scope
- Research paper PDF and mini-project report
- References in a consistent style

Likely viva questions

- Why multi-label instead of single-label? (Posts can express several emotions.)
- Why macro-F1 as the main metric? (Rare emotions count equally; accuracy hides imbalance.)
- Why no stemming or stopword removal? (Multilingual; negations matter.)
- Why char n-grams? (Language-agnostic, robust to spelling and transliteration.)
- Why sigmoid and not softmax? (Independent label probabilities.)
- What is threshold tuning and why on validation data? (Imbalance shifts optimal cutoffs; tuning on test would leak.)
- What does XLM-R add over TF-IDF? (Context and shared multilingual representations.)

14. References (verify details before citing)

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8. E. Saravia et al., CARER: Contextualized Affect Representations for Emotion Recognition, EMNLP 2018.
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11. P. Ekman, An argument for basic emotions, Cognition and Emotion, 1992.
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